COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure Algorithms	**SECTION:**	CpE21S2
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SEATWORK 8.1 Using Sorting Algorithms 2

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using sorting algorithms.

1. To be familiarized with the sorting algorithms.
2. To be able to different the types of sorting algorithms.
3. To be able to create a program with sorting algorithms.

Answer the following questions:

1. Explain the quick sort, shell sort, and merge sort types of sorting algorithms.
    - **Quick Sort** - pivot is in correct position if left side is smaller than pivot and right side is bigger than pivot. Where item from the left side is swapped with item from the right if the left item is bigger than right item. repeat until all items are in the correct position
    - **Merge Sort**- divide the array in half until the array's elements are individual then compare the two individual array and put the larger number to the right and the smaller to the left repeat the process until the array is one again and all elements are in correct position.
    - **Shell Sort** - Similar to insertion sort, but it compares items using a gap. That in which gets smaller after every pass until it becomes 1. Leaving us with a nearly or sorted array that in which can be finished by insertion sort.

2. Give simple sample programs in C++ that use the above sorting algorithms. Use a user input of 10 integer values in your example elements in an array to be sorted. Explain how the programs work.

   - **Quick Sort** can be used in an airline reservation program to arrange flight ticket prices from lowest to highest. It selects a pivot price, then partitions the list so cheaper flights go to one side and pricier ones to the other, repeating this process until all prices are sorted making it ideal for quickly displaying the cheapest available flights first. The system takes in 10 flight prices entered by an agent or pulled from a booking database. Quick Sort picks a price as the "pivot" and rearranges the list so every cheaper flight ends up to its left and every pricier one to its right. This partitioning repeats on each half until the whole list is sorted.
   - **Merge Sort** can be used in a library catalog program to arrange book records (such as book ID numbers) in order, especially when merging records from multiple branches into one master catalog. It splits the book list into smaller pieces, sorts each piece, then merges them back together in the correct order. Real library systems where records from separate branch catalogs (already individually sorted) need to be **merged** into one combined, correctly ordered master list — which is exactly the operation Merge Sort is built around.
   - **Shell Sort** can be used in a warehouse inventory program to arrange item stock quantities in order, helping staff quickly identify which items are low in stock versus overstocked. It compares stock values that are spaced far apart first, gradually shrinking the gap until it behaves like a fast, nearly-finished Insertion Sort. This makes Shell Sort a practical choice for inventory systems with moderate-sized datasets that need reasonably fast sorting without the extra memory overhead of Merge Sort.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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